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CS 307

Operating Systems

Faculty Faculty of Engineering and Natural Sciences

Semester Spring 2025-2026

Course CS 307 - Operating Systems

Time/Place	<u>Time</u>	<u>Week Day</u>	<u>Place</u>	<u>Date</u>
	13:40-14:30	Wed	SBS-G071	Feb 16-May 22, 2026
	14:40-16:30	Thu	SBS-G071	Feb 16-May 22, 2026

Level of course Undergraduate

Course Credits SU Credit:3, ECTS:6, Engineering:6

Prerequisites CS 204

Corequisites CS 307R

Course Type Lecture

Program Requirements >

Instructor(s) Information

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Course Information

Catalog Course Description

This course covers fundamental aspects of operating systems: management and virtualization of resources such as CPU, memory space and peripheral devices. Topics include concurrent processes, mutual exclusion, process communication, cooperation, deadlocks, semaphores, scheduling, protection, memory management, file system abstraction and persistent memory organization. The course will also highlight important aspects of operating systems such as UNIX and Windows.

Course Learning Outcomes:

1.	To have a high-level understanding of computer systems, how an operating system works, and the general structure of an operating system.
2.	To have an understanding of how OS handles multiple processes, and multi-threading including synchronization, scheduling, and memory management.
3.	To be aware of some common problems due to interprocess communication, how these problems are modeled and solved.
4.	To have an understanding of how file systems are implemented and how I/O operations are performed.
5.	To have an understanding of basics of threats to computer security

Course Objective

To supply the students with the theory and practice of operating systems and operating system design principles.

Sustainable Development Goals (SDGs) Related to This Course:

	Affordable and Clean Energy
	Industry, Innovation and Infrastructure

Course Materials

Resources:

Primary Textbook: Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau. Operating Systems: Three Easy Pieces, 1.00 Edition, Published by CreateSpace Independent Publishing Platform, 2018, ISBN: 978-1985086593. Free online access: <https://pages.cs.wisc.edu/~remzi/OSTEP/>

Technology Requirements:

CS307 server for programming assignments:
cs307.sabanciuniv.edu OR 10.3.0.91

Login with your SU username and password.
It will be available after add/drop period.

Assessment

The final grade for this course will be based on:

	Percentage (%)	Number of assessment methods
Final	40%	
Midterm	30%	1
Quiz	14%	7
Assignment	16%	4

Note: This grading scheme is tentative and might change until the add/drop period.

Exam Schedule:

<u>Time</u>	<u>Event</u>	<u>Details</u>
Apr 11, 2026 13:00	Midterm: CS307 - Midterm Exam	

Policies

Course Policies:

1. Grading Policies

1.1 Programming Assignments (PAs)

There will be 4 programming assignments in total. You will be required to write C/C++ programs. Your programs should compile and run properly on UNIX-based operating systems.

For each PA, there will be a related quiz, immediately after the deadline. The quiz score will determine the maximum score you can take from that PA. For each PA, a margin will be announced after the quiz to be added to the quiz score to determine the upper threshold for your PA score. For instance, if your related quiz score is 50 and the announced margin is 20, then the maximum score you can get from this PA is 70.

If you miss a PA quiz and you want your PA to be graded, you have to take an oral exam to determine your PA score.

1.2 Midterm Examination

2-3 hours exam towards the end of 8th week. The midterm exam will be physical and in class. Students will answer both theoretical and computational questions using pen and paper. Exam date will be scheduled by Student Resources (SR) and cannot be modified by course staff.

1.3 Final Examination

2-3 hours exam that will be held during the finals week Exam date will be scheduled by SR and cannot be modified by course staff. The final exam will cover all the topics in the course.

1.4 Quiz

There will be 9 quizzes throughout the semester. Top 7 quiz scores will be counted towards the overall score. Quizzes might take place during lectures or recitations. Quizzes will not be announced beforehand except the PA quizzes. PA quiz scores are counted towards the quiz grading item, too.

There will not be any makeup quizzes.

1.5 Oral Examination

For each of the assessment method above, some students might be randomly called for an oral examination to clarify their work. Depending on the oral exam score, the original assessment score might be modified.

1.6 Overall Score Restriction

Students whose total score is less than 30 or final exam score (or the make-up exam score that will replace the final exam) is less than 25 will automatically fail the course (get an F).

1.6.1 Example Scenario 1

The student scored 30 from the final exam, but the overall score is 20. Then, the student directly fails the course.

1.6.2 Example Scenario 2

The student's overall score is 35, but the final exam score is 20. Then, the student directly fails the course.

1.6.3 Example Scenario 3

The student's both overall score and final exam score are 33. Then, this rule is not conclusive about the letter grade of the student. Depending on the total score distributions, the student might still fail the course.

2. Office Hour Policy

This semester office hours will be conducted online. To attend the office hours of LAs, TAs and the instructor (course staff), students must send an email the corresponding course staff at least a few hours ago. **Last-minute appointment requests will not be accepted.** Each member of the course staff will reserve 2 hours per week for conducting office hours. Office hours are divided into 30-minute slots. Students can request at most one slot per week. Slots will be booked in a first come first served manner. After all the slots become full, course staff **do not guarantee** to open new slots.

3. Email Policy

Course staff are expected to reply to student emails within 48 hours. Course staff might not always provide immediate responses. It is advised that students do not wait until the submission deadline for doing their assignments. Course staff try to be more responsive during the deadline period, but last-minute questions might not be answered due to the heavy email traffic. Students are advised to post their course related questions to the discussion forum in SUCourse since other students might answer them quickly. Course staff has the freedom of not answering a student question if the answer already exists in the course material (syllabus, discussion forums, course book or any other material in SUCourse) or if the student is expected to find the answer herself/himself.

4. Grace Day Policy

Each student has **5 grace days** in total that can be used throughout the semester for extending missed deadlines. It can only be used for extending programming assignment deadlines. The student might use all grace days at once for one assignment or partition them among distinct assignments. Granularity of using grace days is 1-day. For instance, the student cannot ask for a 2 day 10-hour extension and save 2 days 14 hours for later. The student does not have to provide any excuse or document for using his/her grace days. If the student finishes all his/her grace days, no further extension will be given to him/her even if he/she has a valid excuse and documentation.

The student will be able to make submissions/updates on your assignment after the deadline, using your grace days. So, you can use your grace days without notifying us. (i.e. no need to

send us a mail or fill in some sheet).

An excel sheet link is shared with you so that you can see how many grace days you've used for each assignment, as well as how many you have left. You may find the same link on the CS307 Sucourse page under General → Important Information → Grace Day Sheet.

Please note that:

- The system does not keep your submission history, and we cannot access your earlier submissions. Therefore, only your final version will be graded, and final update date will be used to calculate your remaining grace days.
- 'No submission' means you haven't used any of your grace days.
- Even if you don't have any grace days left, the system still technically allows you to upload files after the deadline. However, grace days are manually tracked via the shared sheet. If you submit an assignment exceeding your remaining grace days (as recorded in the sheet), it will be counted as 'No Submission', even if the system accepts your submission. (i.e., if you have 2 grace days left, but you upload more than 2 days after the deadline, you will receive 0.) Therefore, it is your responsibility to check your remaining grace days on the sheet before submitting.

5. Makeup Policy

There is only **one** make-up exam that will be conducted after the final exam. The exam date is decided and announced after the final exam. Only students with a valid excuse (SU health-center approved health report or any other official document) for the midterm or the final exam are eligible for the make-up. See "Student Medical Reports Instruction Letter" for detailed information. The make-up score can replace only one exam. Even though the student has a valid excuse for not attending both final and midterm exams, s/he can take the make-up to replace only one of them.

6. Conflict Policy

Time conflict permissions are not given for lectures or recitations.

7. Academic Integrity & Plagiarism

Cheating and plagiarism will not be tolerated, see Sabanci University's statement on academic integrity for more information.

Students' programming assignment submissions will be compared with other submissions from both the current and previous terms by the course staff using the JPLAG tool. If the similarity score exceeds a predetermined threshold, penalties will be applied. If the JPLAG test fails for the first time, student's score might be reduced to 0 depending on the degree of similarity. For the second JPLAG failure, the obtained score will be -100 for both submissions and for the third failure the student's case will be directly forwarded to the disciplinary committee for further investigation. It will not be possible to regain scores of failed tests back through extra assignments, resubmissions, and oral exams.

7.1. AI Policy

GenAI usage for learning purposes is encouraged. However, copying any AI generated code for PAs and AI generated text for reports are strictly prohibited. Intensive usage of GenAI by many students might cause high similarity among submissions and result in JPLAG failures.

Report submission similarities will also be checked via Turnitin. If the similarity scores exceed a threshold, measures explained for JPLAG failures will be applied.

Note: The instructor holds the right to decide a policy concerning issues not already covered here.

AI Policies

[Guidelines on the "Use of Generative AI" for students](#)

SU Academic Integrity Statement:

<https://www.sabanciuniv.edu/en/academic-integrity-statement>

Health Report Requirements

[Student Medical Reports Instruction Letter](#) 

Weekly Content

Week	Topic
Week 1	Introduction
Week 2	CPU Virtualization: Processes and System Calls
Week 3	CPU Virtualization: Limited Direct Execution
Week 4	Scheduling: Basic Algorithms
Week 5	Scheduling: Proportional Share and Realistic Algorithms
Week 6	Concurrency: Threads & Mutexes
Week 7	Concurrency Primitives: Condition Variables and Semaphores
Week 8	Concurrency: Deadlocks and Concurrency Errors
Week 9	Memory Virtualization: Virtual Address Spaces
Week 10	Memory Virtualization: Address Translation Mechanisms and Free Space Management

Week	Topic
Week 11	Memory Virtualization: Swap Space
Week 12	Persistence: Peripheral Devices and I/O Handling
Week 13	Persistence: File System API and Implementation Problems
Week 14	Security: Authentication & Access Control Mechanisms